

KUSTAVI ISOKARI, FinlandWMO: **029640**Lat: **60.722N**Lon: **21.027E**Elev: **13**StdP: **14.69**Time Zone: **2.00 (EUE)**Period: **04-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	3.7	9.9	-0.4	5.4	4.1	5.2	7.2	11.2	38.6	32.1	35.9	32.0	12.5	110	0.830

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.1	73.7	67.4	71.1	65.4	68.7	63.7	68.7	72.5	66.5	69.8	64.6	68.1	12.0	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.2	99.9	71.2	65.1	92.9	68.9	63.0	86.2	66.9	32.9	72.8	31.1	69.8	29.7	68.1	74.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 35.9	31.9	28.8	4.3	76.6	8.1	3.4	-1.5	79.0	-6.2	81.0	-10.8	82.9	-16.6	85.4		
(5)			2.6	68.5	6.9	2.8	-2.4	70.6	-6.4	72.2	-10.3	73.8	-15.3	75.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	44.1	28.6	26.7	30.2	37.4	47.2	55.4	63.1	63.1	56.3	46.5	39.6	34.3
	DBStd	13.87	7.80	8.99	6.26	4.06	5.23	4.79	4.73	4.13	4.29	5.00	5.47	7.06
	HDD50	3365	663	652	614	378	118	9	0	0	6	128	311	486
	HDD65	7678	1128	1072	1079	827	552	291	95	86	263	573	761	951
	CDD50	1226	0	0	0	1	32	169	405	405	194	20	0	0
	CDD65	64	0	0	0	0	0	2	36	26	0	0	0	0
	CDH74	73	0	0	0	0	0	2	48	23	0	0	0	0
	CDH80	1	0	0	0	0	0	0	1	0	0	0	0	0
	WSAvg	15.4	17.5	15.5	15.1	14.1	13.2	13.1	12.7	13.9	15.4	17.1	18.5	18.9
	PrecAvg	23.0	2.0	1.4	1.2	1.1	1.3	1.8	2.2	2.6	2.2	2.6	2.2	2.2
(15) Precipitation	PrecMax	28.6	3.6	3.4	3.0	2.2	3.5	3.7	4.8	5.7	5.6	5.8	5.4	5.5
	PrecMin	17.0	0.2	0.1	0.1	0.0	0.2	0.4	0.0	0.5	0.3	0.2	0.4	0.2
	PrecStd	3.2	0.9	0.8	0.6	0.6	0.7	0.8	1.3	1.4	1.6	1.3	1.1	1.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.2	39.5	42.4	53.7	66.6	71.5	78.3	77.4	67.0	57.9	50.2	46.0
		MCWB	39.9	37.1	38.5	46.6	55.6	62.3	69.5	70.1	62.4	55.8	48.8	44.4
	2%	DB	39.8	37.7	40.0	48.0	60.2	67.1	74.8	73.1	64.5	56.0	48.8	44.1
		MCWB	38.6	36.0	37.1	42.8	53.6	59.6	67.9	67.2	60.6	53.9	47.6	42.4
	5%	DB	38.6	36.6	38.4	45.0	56.8	64.2	72.2	70.6	63.0	54.4	47.7	42.8
		MCWB	37.5	35.4	36.2	41.2	51.9	58.5	66.8	64.9	59.6	52.3	46.3	41.2
	10%	DB	37.3	35.5	37.1	42.5	54.2	61.8	69.7	68.5	61.6	53.0	46.1	41.4
		MCWB	36.1	34.7	35.4	39.6	50.0	57.0	64.8	63.5	58.6	51.0	44.5	39.9
	0.4%	WB	40.4	37.8	39.5	47.1	58.2	64.0	72.5	72.1	63.8	56.7	49.4	45.0
		MCDB	40.7	38.6	41.8	52.5	63.3	69.9	75.8	75.1	66.2	57.5	49.6	45.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	39.1	36.4	37.7	43.7	54.8	61.1	69.8	68.3	61.8	54.4	48.0	43.1
		MCDB	39.5	37.3	39.2	47.2	59.0	65.3	73.4	72.2	63.8	55.5	48.8	43.8
	5%	WB	37.7	35.6	36.7	41.8	52.5	59.3	67.4	66.2	60.3	53.1	46.6	41.4
		MCDB	38.4	36.4	38.0	44.5	56.3	63.0	70.9	69.0	62.5	54.3	47.7	42.4
	10%	WB	36.3	34.7	35.7	40.1	50.5	57.7	65.7	64.4	59.2	51.5	44.9	40.2
		MCDB	37.2	35.5	37.0	42.2	53.7	60.9	69.3	67.6	61.4	52.8	45.8	41.2
(35) Mean Daily Temperature Range	5% DB	MDBR	5.7	5.9	6.5	6.9	8.1	7.5	6.9	6.1	5.2	4.5	4.2	4.7
		MCDBR	4.5	4.1	6.1	10.6	13.2	11.7	9.8	7.9	6.3	4.5	3.8	4.6
		MCWBR	4.3	4.0	4.6	6.4	7.3	6.0	5.0	4.5	4.3	4.5	4.1	4.7
	5% WB	MCDBR	4.4	4.1	5.9	9.9	12.6	10.2	8.8	6.8	5.9	4.5	3.7	4.3
		MCWBR	4.5	3.9	4.7	6.3	7.5	6.0	5.0	4.7	4.5	4.8	4.2	4.6
(40) Clear-Sky Solar Irradiance	taub		0.272	0.311	0.329	0.368	0.367	0.357	0.386	0.374	0.347	0.330	0.295	0.374
	taud		2.164	2.248	2.308	2.322	2.416	2.483	2.430	2.450	2.509	2.406	2.206	2.592
	Ebn at Noon		148	213	251	262	272	277	265	259	247	206	136	101
	Edh at Noon		13	21	28	33	32	31	32	29	23	18	12	9
(44) All-Sky Solar Radiation	RadAvg		69	261	751	1264	1743	1924	1779	1383	834	352	93	42
	RadStd		10	48	85	118	144	127	162	121	92	54	18	5

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (46 neighbors)		+0.72	N/A	N/A	N/A	N/A	N/A	-214	N/A	N/A	N/A	

Nomenclature: See separate page